Maximum compactness of static stars when the lapse is tied to the isotropic conformal factor, N = 2/ψ − 1

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Abstract

In isotropic coordinates the Schwarzschild metric satisfies N = 2/ψ − 1 for every mass, so if the lapse is a function of the isotropic conformal factor alone, vacuum general relativity fixes that function (in areal coordinates the same argument gives a different closure, Jacobson 2007: the choice of the isotropic variable is an assumption). Imposing this closure inside matter, while keeping the Hamiltonian constraint and energy–momentum conservation of a perfect fluid, gives a phenomenological closure of the static equations in which the fluid pressure does not enter the field equations. Read with Einstein's equations, the stars are anisotropic. For uniform density, in closed form: maximum compactness 2^(1/3) − 2^(−1/3) = 0.4662 against Buchdahl's 4/9, surface redshift 2.847 against 2, and 0.4043 against 3/8 if p_c ≤ ρ is imposed. For five density profiles the maximum compactness is 3–5% above GR. At observed compactness the required central pressure is already lower than in GR (by 7% at C = 0.15), so realistic mass–radius curves differ. With the SLy and APR4 equations of state the closure raises the maximum mass by about 5% and the radius of a 1.4 solar-mass star by 0.1 km, about ten times less than present NICER errors. GR and the closure are both compatible with PSR J0437−4715; at 2.08 solar masses both give radii below the 68% interval of Salmi et al. (2024) for PSR J0740+6620. With current data the closure cannot be distinguished from general relativity.

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Version 2.34.2 (October 2026). This version corrects some figures of version 2.34.1 (binding energy at exact mass, the reference value of ALF2, a numerical noise in the radii of the closure stars, the NICER comparison) and adds the checks made for a journal manuscript: closed forms of the effective stress, a bound C < 12/25 for non-increasing densities, and the post-Newtonian expansion of the interior. Earlier sections are unchanged; see «Added in version 2.34.2».

Version 2.34.1 (October 2026). This version adds the prior work on the closure, which version 2.34 did not cite: the relation N = 2/ψ − 1 inside matter is the assumption made by Buchdahl (1964) for his n = 5 solution. It also adds two checks: maximum compactness with causal equations of state, and the binding energy of neutron stars. The programs, results and notes of version 2.34 are unchanged; the additions are in the section «Added in version 2.34.1».

In brief

In general relativity (GR) a static star made of a perfect fluid, with density not increasing outwards, cannot be more compact than 4/9, in units GM/(Rc²) (Buchdahl 1959). This deposit studies one alternative rule for the inside of a static, spherical star. It ties the lapse N (the rate of clocks) to the isotropic conformal factor ψ of space:

N = 2/ψ − 1.

For a single spherical body in vacuum this relation holds exactly: it is a property of the Schwarzschild metric in isotropic coordinates. Imposing it inside matter as well changes one thing in the field equations: pressure no longer acts as a source of the lapse. Read with Einstein's equations, the resulting metric then corresponds to the fluid plus an extra anisotropic effective stress. The differences from GR inside matter are of relative order p/ρ, so they are negligible in ordinary stars and appear only in neutron stars. The results:

- for uniform density the sequence of stars ends at a compactness 2^(1/3) − 2^(−1/3) ≈ 0.466, in closed form, instead of 4/9 ≈ 0.444 (for density decreasing outwards the maximum is lower; see the table below);
- for uniform density, at compactness between 0.15 and 0.23, the central pressure needed is 7–12% lower than in GR;
- with two realistic nuclear equations of state the maximum mass rises by about 5% and radii change by about 0.1 km, below present NICER errors. With these two equations of state GR and the closure are compatible with PSR J0437−4715, and both give radii below the 68% interval of Salmi et al. for PSR J0740+6620.

The rule is an assumption, not a derivation, and this is not yet a theory: there is no action and no dynamics.

Why this rule

In vacuum, for a single body, the lapse is a function of the isotropic conformal factor alone: $N = 2/\psi - 1$. Since the Schwarzschild metric depends on the mass only through M/r, the same function holds for every M. The assumption of this deposit is that this functional relation also holds inside matter. In GR the Hamiltonian constraint sources ψ with the energy density only, while pressure enters the equation for the lapse; imposing $N = N(\psi)$ therefore removes pressure as a source of the lapse; the Hamiltonian constraint and the fluid equations are unchanged.

The rule was suggested to the author by an informal picture in which the lapse at a place depends only on the conformal factor there, not on the internal stresses of matter. It is explained, for readers who want it, in `WHERE_THIS_COMES_FROM.md`; nothing in the calculations depends on it.

Neutron stars are the only static objects in which pressure is comparable to energy density, so they are where this rule can be tested.

The equations

Static, spherically symmetric, isotropic coordinates: $ds^2 = -N^2dt^2 + \psi^4(dr^2 + r^2d\Omega^2)$, $G = c = 1$, ρ = total energy density measured by static observers.

1. **Hamiltonian constraint, as in general relativity (GR):** $\nabla^2\psi = -2\pi\rho\psi^5$.
2. **The closure, replacing GR's lapse equation $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$:** $N = 2/\psi - 1$, i.e. $\psi + N\psi = 2$.
3. **Matter:** a perfect fluid obeying $\nabla_\mu T\^\mu\nu = 0$, i.e. $p' = -(\rho + p)N'/N$, with $p = 0$ at the surface.

Where the closure comes from, and what is assumed

In isotropic coordinates the Schwarzschild metric has $\psi = 1 + M/2r$ and $N = (1 - M/2r)/(1 + M/2r)$, so $N = 2/\psi - 1$ for every M. If the lapse is a function of the isotropic conformal factor alone, vacuum GR fixes that function uniquely.

The choice of variable matters. In areal coordinates the same argument gives $N^2g_RR = 1$ (Jacobson 2007). Inside matter that relation forces $p_r = -\rho$ and allows compactness up to 1/2. The results below are a property of the **isotropic** closure. The isotropic variable is the author's choice, motivated in `WHERE_THIS_COMES_FROM.md`; it is an assumption: the spatial metric is taken to be conformally flat with ψ as the physical variable, and the closure is imposed inside matter too. ψ is obtained by integrating the constraint over the whole star, so the closure is a pointwise relation between two fields, not a local equation of state.

One body only. The statement is about exact spherical symmetry. For $\psi = 1 + a_1/|x - x_1| + a_2/|x - x_2|$ (harmonic; the Brill–Lindquist conformal factor of time-symmetric vacuum data) with $N = 2/\psi - 1$, the static metric is not a vacuum solution: its Ricci tensor is non-zero at order a_1a_2 in an open region, and vanishes when either mass is removed (`codice/domini_e_masse.py`).

What it implies

The fluid pressure does not enter the field equations: the metric, and hence its effective stress, is fixed by ρ alone. Read with Einstein's equations, the same metric has an anisotropic effective stress ($p_r \neq p_t$, with $p_t > p_r$). Already in the Newtonian limit the effective radial stress at the centre is half the fluid pressure. The difference between the effective and the fluid stress tensors is conserved and has zero energy density. **This is a phenomenological closure of the static equations, not yet a theory:** there is no action and no dynamics.

Local versus global. With $\psi \to 1 + M_ADM/2r$ and $N \to 1 - M_Komar/r$, the flux of the flat gradient gives $\int\nabla^2(\psi + N\psi)\, d^3x = 4\pi(M_Komar - M_ADM)$. In GR the integrand, $2\pi\psi^5[N(\rho + 6p) - \rho]$, is non-zero inside matter but integrates to zero, because Komar and ADM masses coincide for stationary asymptotically flat spacetimes (Beig 1978). The closure sets this integrand to zero pointwise (`codice/domini_e_masse.py`).

Related theories

Other proposals also leave vacuum GR unchanged (up to a cosmological constant) and modify only the interior of matter: Eddington-inspired Born–Infeld gravity (Bañados and Ferreira 2010), whose neutron stars were studied by Pani,

Cardoso and Delsate (2011), and Palatini f(R) gravity (Olmo 2011). They differ from the present rule in two ways. They follow from an action, which the rule does not have. And their interiors depend on a coupling (κ in Born–Infeld gravity, the function f in Palatini f(R)), whereas here, once the isotropic variable is chosen (an assumption, stated above), the relation between N and ψ inside matter is the one fixed by the vacuum solution. The rule itself has no coupling constant; the equation of state remains an input. For uniform density this gives the single value 0.466.

Where the differences come from

In these variables ψ is sourced by the energy density alone in GR and in the closure, through the same Hamiltonian constraint. The only difference is the equation for the lapse. In GR pressure is a source of the lapse: covariantly $D^2N = 4\pi N(\rho + 3p)$, which in the flat-Laplacian form used above reads $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$. Here the lapse follows from ψ, so pressure does not source it. With less gravitational pull from pressure, the same star needs less central pressure to be in equilibrium, and a given equation of state supports a little more mass.

The two sequences end for different reasons. In GR the sequence of uniform-density stars ends at 4/9, where the central pressure required for equilibrium diverges. Here it ends where the central lapse reaches zero, ψ_c = 2 (the pressure also diverges there), at 0.466.

The effect is of relative order p/ρ. It is present in any matter, but in the Sun or the Earth p/ρ is tiny and GR and the closure agree to that accuracy. Only in neutron stars, where p/ρ reaches tens of per cent, do they separate measurably.

Where the sequence of stars ends

For uniform density, (ρ + p)N is constant in both GR and this closure, so the central pressure diverges exactly when the central lapse vanishes. In GR pressure is a source of the lapse (ρ + 6p in the flat-Laplacian form), so pressure feeds back on the geometry. Here N is fixed by ψ, which depends on ρ alone. The sequence ends at ψ_c = 2, the edge of the range where $N = 2/\psi - 1 \geq 0$.

Uniform density, in closed form. The constraint is the n = 5 Lane–Emden equation, solved by $\psi = \psi_c(1 + 2\pi\rho\psi_c^4 r^2/3)^{(-1/2)}$. Matching to the exterior gives $\psi_R = \psi_c^{(2/3)}$, and the compactness is $C = GM/(Rc^2) = 2(\psi_R - 1)/\psi_R^2$, with R the areal radius.

- GR's Buchdahl limit, C = 4/9, corresponds to ψ_R = 3/2 (ψ_c = 1.8371).
- The limit ψ_c = 2 gives **$C_{max} = 2^{(1/3)} - 2^{(-1/3)} = 0.46622$**. Equivalently, the surface redshift is z = 2.847 (2 in GR). This is below Ivanov's bounds for anisotropic stars (3.842 or 5.211, depending on the energy condition imposed on the tangential pressure).
- Requiring the fluid pressure to satisfy p_c ≤ ρ gives $y^3 + 2y - 4 = 0$ with $y = \psi_c^{(1/3)}$, i.e. C = 0.40427 (3/8 in GR). This holds for uniform density only.
- The effective radial stress at the centre is $p_r/\rho = (\psi_c - 1)/(3(2 - \psi_c))$.

Relation to Buchdahl's hypotheses. Andréasson's theorem (2008) states that a static, spherically symmetric GR solution with p_r ≥ 0 and p_r + 2p_t ≤ ρ has 2m/r ≤ 8/9, with no requirement on how the density varies. Here p_r ≥ 0, so exceeding 4/9 requires violating the trace condition. That happens at the centre for ψ_c > 3/2 (C > 0.3615). GR's own uniform star violates the same condition for C > 5/18. What takes the configuration outside Buchdahl's hypotheses is the anisotropy. The dominant energy condition for the effective stress fails for ψ_c > 7/4 (C > 0.4289), but that is not needed to exceed 4/9.

Other density profiles (numerical). ρ(r)/ρ_c = f(x), with x = r/R and r the **isotropic** radius. These values are not comparable with literature values for profiles of the same name, which use the areal radius. Compare within a row. The last digit is ±0.0005.

f(x)	max compactness, isotropic closure	GR, same profile and coordinate	ratio
1	0.4662	0.4444	1.049
$1 - x^2$	0.4253	0.4052	1.049
$1 - x$	0.4084	0.3922	1.041
$(1 - x^2)^2$	0.3937	0.3750	1.050

$(1-x)^2$	0.3573	0.3457	1.034

The maximum compactness is 3–5% higher than in GR for every profile tested. Among these profiles the uniform one is the most compact. This is not proved over all non-increasing profiles.

Neutron stars below the limit (uniform density)

For the same profile and the same ψ_c, the spatial geometry, and hence M and R, is the same as in GR. The lapse and the pressure are not. For uniform density the required central pressure is lower than in GR:

C	**0.15**	**0.23**	**0.29**	**0.34**	**0.375**
p_c (closure) / p_c (GR)	0.93	0.88	0.82	0.75	0.67

Realistic equations of state and NICER

How the equation of state enters. The same three equations as above, with a realistic equation of state $\varepsilon(p)$ for cold nuclear matter (ε is the total energy density, called ρ above): ψ from the Hamiltonian constraint $\nabla^2\psi = -2\pi\varepsilon\psi^5$ (ε the total energy density), $N = 2/\psi - 1$, and p from conservation, $p' = -(\varepsilon + p)N'/N$. The central value ψ_c is fixed by $\psi \to 1$ at infinity; the mass is read from the exterior $\psi = 1 + M/2r$ and the areal radius is $\psi(R)^2 R$. The comparison is GR (the TOV equations) with the same equation of state. We use the piecewise-polytrope forms of SLy and APR4 of Read et al. (2009), with their SLy crust. Stars are counted on the branch with $dM/dp_c > 0$; for the closure this static criterion is an assumption, since there is no dynamics. Six predictions for this comparison were written down, with a SHA-256 fingerprint of the file, before computing (`documenti/PREVISIONI_MASSA_RAGGIO.md`): (1) the GR values reproduce Read et al. within 2%; (2) the closure raises the maximum mass by at least 3%; (3) it shifts the radius at 1.4 $M_\odot$ by less than 0.5 km, (4) upwards; (5) GR and the closure both fall in the PSR J0437−4715 interval; (6) both reach at least 2.01 $M_\odot$. All six held. This section has not yet been re-run independently.

EOS	**theory**	**M_{max} ($M_\odot$)**	**R at 1.4 $M_\odot$**	**R at 1.418 $M_\odot$**	**R at 2.08 $M_\odot$**
SLy	GR	2.048	11.71 km	11.69 km	— (not reached)
SLy	closure	2.146	11.84 km	11.83 km	10.83 km
APR4	GR	2.188	11.33 km	11.33 km	10.68 km
APR4	closure	2.298	11.43 km	11.42 km	11.03 km

The GR values agree with Read et al. within 1.1% (their fitted piecewise forms differ from the tabulated equations of state by the same amount). Figure: `risultati/massa_raggio.png`.

Against NICER (68% intervals). PSR J0437−4715: $M = 1.418 \pm 0.037$, $R = 11.36\ (+0.95/-0.63)$ km (Choudhury et al. 2024). All four curves fall inside it. PSR J0740+6620: $M = 2.08 \pm 0.07$, $R = 12.49\ (+1.28/-0.88)$ km (Salmi et al. 2024; Dittmann et al. 2024 give 12.92 (+2.09/−1.13) km). GR and the closure, with these two equations of state, give radii at 2.08 $M_\odot$ below the Salmi et al. interval; the closure is closer by about 0.35 km (APR4).

What this means. For the same equation of state the closure raises the maximum mass by about 5% and moves the radius of a 1.4 $M_\odot$ star by 0.10–0.13 km. That radius shift is about ten times smaller than the present NICER errors (about ±0.8 km), so **current data do not distinguish the closure from GR**. The 5% in maximum mass is degenerate with the stiffness of the equation of state: a slightly stiffer equation of state in GR gives the same. Separating the two would need radius errors of order 0.1 km or an independent determination of the equation of state. With SLy and APR4 the closure is compatible with PSR J0437−4715 and, like GR, gives radii below the Salmi et al. interval for PSR J0740+6620; it is not confirmed by these data.

What is not claimed

- That the closure holds inside matter: it is assumed.
- That the isotropic variable is preferred over others: this is assumed too (see above).

- A covariant or dynamical formulation, so nothing on radial stability, rotation or post-Newtonian constraints on a dynamical completion.
- A bound over all density profiles or equations of state.
- Causality: for uniform density the sound speed is infinite in both GR and this closure.
- That NICER data favour the closure over GR: with SLy and APR4 the two are indistinguishable within present errors.

Added in version 2.34.1

Prior work: Buchdahl's 1964 assumption

The relation N = 2/ψ − 1 inside matter is not new.

- **Buchdahl's choice.** Looking for a relativistic analogue of the Emden polytrope of index 5, Buchdahl (1964) chose isotropic coordinates with ψ = 1 + f and N = (1 − f)/(1 + f) (his eq. 2.2), that is N = 2/ψ − 1. His reason was the one given above: the exterior Schwarzschild metric has this form (his eq. 2.3).
- **His derivation.** He then required a perfect fluid. The isotropy of the pressure gives f f″ − 3f′² − f f′/r = 0 (his eq. 2.7), whose solution is f = (A + Br²)^(−1/2) (his eq. 2.8).
- A = 0 gives the Schwarzschild exterior.
- A > 0 gives his n = 5 solution, which has infinite radius and finite mass.

The program `codice/buchdahl_1964_isotropia.py` redoes this calculation. Version 2.34 cited Buchdahl (1959) for the 4/9 bound, but not this paper.

What this deposit does differently. Buchdahl requires the metric to be a perfect-fluid solution of Einstein's equations, which leaves only his n = 5 family. Here the relation replaces the lapse equation for any density profile:

- the matter is still modelled as a perfect fluid, whose pressure follows from conservation;
- the stars have a finite radius;
- what is dropped is the isotropy of the effective stress obtained by reading the metric with Einstein's equations: that effective stress is anisotropic.

The closed-form maximum 2^(1/3) − 2^(−1/3) for uniform density, the five profiles and the realistic equations of state are consequences of this second choice.

Other prior work found by the novelty check of 7 October 2026 (`documenti/novita/NOVITA_ESITO.md`, in Italian):

- the metric of Buchdahl's solution in the form ψ = 1 + f is entry «Buch2» of the catalogue of Delgaty and Lake (1998);
- the conformal factor of the uniform-density star is the n = 5 Lane–Emden solution of the Hamiltonian constraint, studied as initial data, without a lapse, by Baumgarte, Ó Murchadha and Pfeiffer (2007);
- Puttasiddappa, Bretón and Pérez Bergliaffa (2026) build uniform-density anisotropic stars, also beyond Buchdahl's bound, with the anisotropy proportional to ρ k^μ∇_μP_r through a constant coefficient. The stars of this deposit are not in that family: the corresponding ratio varies along the radius (`codice/confronto_put.py`).

Causal equations of state and the turning point

- **Registered predictions:** `documenti/causalita_stabilita/PREVISIONI.md` (SHA-256 826bfd73…, 7 October 2026).
- **Program and output:** `codice/causale.py`, output `risultati/causale.log`.
- **Equation of state:** p = c_s²(ε − ε_s). c_s² = 1 is the stiffest causal one.
- **Stability:** judged only by the turning point (the maximum of the mass along the sequence). For barotropic stars in GR this criterion is exact; the closure has no dynamics, so for it the criterion is only a proxy.

c_s^2	compactness at maximum mass, GR	compactness at maximum mass, closure	maximum mass, closure/GR
1	0.3527	0.3663	+6.2 %
1/2	0.3040	0.3174	+5.6 %

- **Code check.** The GR value 0.3527 matches the 0.354 of the literature (Koranda, Stergioulas and Friedman 1997) within the resolution of the grid of central pressures.
- **Compactness.** With the stiffest causal equation of state the closure reaches 0.366 at maximum mass. This is below the 0.389 of Alho, Natário, Pani and Raposo (2022) for causal configurations that are also radially stable, and below their 0.462 for causal ones. The value 0.466 remains the incompressible one, to be compared with Buchdahl's 4/9.
- **Energy conditions.** Read with Einstein's equations, the maximum-mass closure star with $c_s^2 = 1$ satisfies the dominant energy condition everywhere.
- **Outcome.** Five registered predictions out of five held.
- **Limits.** Causality here refers to the fluid ($dp/d\varepsilon \leq 1$). The effective medium read in GR has no defined wave speeds without a model of the anisotropy, and a true radial-stability analysis would need a dynamics.

Binding energy of neutron stars

- **Registered predictions:** `documenti/energia_legame/PREVISIONI_ENERGIA_LEGAME.md` (SHA-256 80a36072…, 10 October 2026).
- **Programs:** `codice/energia_legame.py` and `codice/ricontrollo_masse.py`. The second was written after the calculation and is not registered. Outputs are in `risultati/`.
- **Equations of state:** the 17 piecewise polytropes of Read et al. (2009). Their parameters were typed from memory, so they were checked against the published maximum masses: all 17 agree within 1.1 %.

Results.

- **At fixed mass the binding energy $BE = M_B - M$ is nearly the same in both theories.**
- Between 1.25 and 1.6 $M_\odot$ the closure equals GR within 0.1 %.
- At the GR maximum mass it is lower by 0.4–0.75 %, although the radius there grows by 0.61–1.18 km.
- A likely reason is that both theories share the Hamiltonian constraint, which fixes the spatial geometry from the energy density; this is a reading of the numbers, not a proof.
- **The quasi-universal relation $BE/M(\beta)$ does not distinguish the closure.** The mean offset is +2.4 %, against a GR scatter of 4.3 %.
- **PSR J0737−3039B** (Podsiadlowski et al. 2005) selects the same five equations of state in both theories.

Two predictions did not hold (registered text translated from Italian, then the outcome).

- **E2** (a check in GR): «the fit $BE/M = d_1\beta + d_2\beta^2$ over all stable stars of the admitted equations of state has d_1 between 0.55 and 0.70 and a relative (rms) scatter below 5 %». Outcome: $d_1 = 0.480$ and $d_2 = 0.634$, with scatter 4.34 % (505 stars); the scatter holds, d_1 does not.
- **E3** (same equation of state, same mass $M = 1.2489\ M_\odot$): «the closure gives a lower binding energy than GR, between 0.5 % and 5 % less, for every admitted equation of state». Outcome: the differences lie between −0.10 % and +0.02 %.

A caveat on the absolute values. The registered range for d_1 was taken from published fits (Lattimer and Prakash 2001; Breu and Rezzolla 2016), so the outcome of E2 means that our GR binding energies are lower than those fits. A possible cause is the unit of baryon mass; it is not isolated here. This does not affect the comparison between the two theories, which use the same definition. It does affect which equations of state are compatible with J0737−3039B.

Three registered predictions out of five held.

What changes in what is claimed

- **The relation is Buchdahl's.** What this deposit proposes is its use as a closure for any density, and the consequences of that use.
- **Causal equations of state.** Using the turning point as a proxy, the closure reaches $C = 0.366$ at maximum mass, below the cited benchmarks 0.389 (causal and radially stable configurations) and 0.462 (causal configurations). The radial stability of the closure has not been established.
- **With present uncertainties on the equation of state, the static observables tested do not separate the two theories.**
- The maximum mass (+3.7 to +5.3 % for the 17 realistic equations of state, +5.0 to +6.2 % for the causal ones above) and the radius near the maximum mass remain degenerate with the equation of state.
- The binding energy does not separate them either.

- More precise mass–radius measurements (radii to about 0.1 km) or an independently constrained equation of state could help. Tidal deformability and oscillations would also be useful, but the closure has no dynamics to compute them.

Added in version 2.34.2

These are the checks made while preparing a journal manuscript on this closure. They correct some figures of version 2.34.1. The earlier sections are left as they were; where they differ from this section, this section holds.

Corrections

1. **Binding energy at fixed mass.** Version 2.34.1 interpolated the binding energy on a grid of stars. Evaluated at the exact gravitational mass (`codice/legame_massa_esatta.py`), the closure value is lower than the GR value for every admitted equation of state:
2. by 0.02–0.05 % at 1.2489 M☉, and by 0.03–0.17 % at 1.6 M☉ (version 2.34.1: −0.10 to +0.02 % and −0.15 to +0.02 %);
3. at the GR maximum mass (`codice/legame_massa_massima.py`), by 0.43–0.80 %, with radii larger by 0.74–1.36 km (version 2.34.1: 0.4–0.75 % and 0.61–1.18 km).

The registered prediction E3 (0.5–5 % lower at 1.2489 M☉) still does not hold. The sentence «the closure equals GR within 0.1 % between 1.25 and 1.6 M☉» becomes «lower by less than 0.2 %».

1. **ALF2.** `codice/energia_legame.py` uses 1.991 M☉ as the reference maximum mass of ALF2. Table III of Read et al. (2009) gives 2.086 M☉. With the correct value, the registered admission rule (GR maximum mass within 2 % of the tabulated one) excludes ALF2: our GR value, 1.976 M☉, reproduces their piecewise fit, which itself differs by −5.3 % from the tabulated equation of state.
2. The admitted equations of state are 16, not 17 (`codice/energia_legame_senza_ALF2.py`).
3. Unchanged without ALF2: the range of the maximum-mass increase (+3.7 to +5.3 %), the mean offset of the closure from the GR fit BE/M(β) (+2.4 %, against a scatter of 4.4 % about our GR fit), and the five equations of state selected by PSR J0737−3039B in both theories.
4. `codice/energia_legame.py` and its results are left as deposited.
5. **Radii of the closure stars.** With the settings of `codice/massa_raggio.py` (relative tolerance 10^{-9}, maximum step 0.05 km) the radius of the closure stars carries a numerical noise of about 0.01 km, up to 0.03 km, which comes from the crust (`codice/rumore_raggio.py`). With tight settings (`codice/massa_raggio_stretto.py`: 10^{-11}, maximum step 5 m) the curves are smooth to about 0.001 km. Masses are unchanged.

EOS	theory	M_max (M☉)	R at 1.4 M☉	R at 1.418 M☉	R at 2.073 M☉
SLy	GR	2.0485	11.709 km	11.698 km	— (not reached)
SLy	closure	2.1456	11.831 km	11.823 km	10.864 km
APR4	GR	2.1881	11.321 km	11.317 km	10.700 km
APR4	closure	2.2980	11.422 km	11.420 km	11.039 km

The radius shift at 1.4 M☉ is 0.12 km (SLy) and 0.10 km (APR4), not 0.13 and 0.10 km. Figure: `risultati/fig3_mass_radius.pdf`.

1. **NICER.**
2. The mass 1.418 ± 0.037 M☉ for PSR J0437−4715 is the posterior of Choudhury et al. (2024).
3. For PSR J0740+6620, Salmi et al. (2024) give 2.073 ± 0.069 M☉; 2.08 ± 0.07 M☉ is the radio-timing mass used as their prior. The conclusions do not change: radii below their interval, and SLy in GR does not reach this mass.
4. A later analysis of PSR J0437−4715 that includes a modulated non-thermal component (Miller et al. 2026) gives 11.9–15.5 km at 68 %. All four curves lie below it: 11.32–11.82 km at 1.418 M☉.
5. **APR4 and MPA1 are acausal near the maximum mass.** At the centre of the maximum-mass stars the piecewise fit has $dp/d\varepsilon = 1.38$ (GR) and 1.31 (closure) for APR4, and 1.09 and 1.04 for MPA1; for SLy the GR value there is 1.00, at the causal limit (`codice/velocita_centro.py`). The upper end of these sequences lies outside the causal range of the fit.
6. **Causal equations of state.** The maximum of the mass is now located precisely, not read on the grid

(`codice/causale_massimo.py`):

c_s^2	compactness at maximum mass, GR	compactness at maximum mass, closure	maximum mass, closure/GR
1	0.3540	0.3668	+6.2 %
1/2	0.3060	0.3157	+5.6 %
1/3	0.2707	0.2784	+5.0 %

- The GR value for $c_s^2 = 1$ agrees with the 0.354 of Lattimer and Prakash (2016).
- The conjecture that this equation of state gives the most compact stars is stated in their formulation, based on Koranda, Stergioulas and Friedman (1997); version 2.34.1 attributed the number to the latter.
- **A sentence withdrawn.** Version 2.34.1 wrote that «a slightly stiffer equation of state in GR gives the same» maximum mass as the closure. This was not computed. What remains is that the increase is of the same size as the differences between equations of state.

New results

Effective stress in closed form (`codice/traccia_sforzo.py`, symbolic). Read with Einstein's equations, every closure configuration has, for any density profile:

- $\rho_{eff} = \rho$;
- trace $S = p_r + 2p_t = \rho(\psi - 1)/(2 - \psi)$;
- $p_r = -\psi'[(r\psi)' - 1]/[2\pi r\psi^5(2 - \psi)] \geq 0$.

The trace condition $p_r + 2p_t \leq \rho$ holds exactly where $\psi \leq 3/2$.

A bound for non-increasing densities. Every closure configuration whose density does not increase outwards has $C < 12/25 = 0.48$. The proof takes three steps:

1. $q = \rho\psi^5$ does not increase, so $\tilde{m}(r)/r^3$, with $\tilde{m} = -r^2\psi'$, does not increase;
2. hence $\psi_c - \psi_R \geq (\psi_R - 1)/2$;
3. positivity of the central lapse, $\psi_c < 2$, then gives $\psi_R < 5/3$.

Context:

- **Initial data.** The bound uses only the Hamiltonian constraint and $\psi_c < 2$.
- **Without monotonicity** thin shells approach $C = 1/2$, that is $2M/R \to 1$.
- **Equation of state.** For stars built from a barotropic equation of state the density is automatically non-increasing.
- **Numerical checks.** No step profile exceeds the uniform value $2^{(1/3)} - 2^{(-1/3)} = 0.46622$, which remains a conjecture for the supremum:
- 120 random non-increasing step profiles (`codice/profili_a_gradini.py`): the largest is 0.4662163;
- a local optimisation over two-step profiles returns the uniform one (`codice/gradini_ottimizzazione.py`, written independently by an AI assistant acting as referee).

Post-Newtonian interior (`codice/ppn_interno.py`, `codice/phi_w_sfera.py`). To post-Newtonian order the spatial metric is the same in both theories; the lapse gives

- $-g_{00} = 1 - 2U + 2U^2 - 4\Phi_2 - 6\Phi_4$ in GR;
- $-g_{00} = 1 - 2U + 2U^2 - 5\Phi_2$ with the closure.

The second line can be read in two ways.

- **Read within GR:** the metric is that of GR with an effective stress whose trace is $S = \rho U/2$ to leading order. All post-Newtonian parameters are those of GR, and the closure is a statement about matter. That matter is nonlocal: through ψ, the stress at a point depends on the density of the whole configuration, inside and outside r, and on the normalization of ψ at infinity. This is a stronger nonlocality than in the equations of state of Hernández and Núñez (2004), which depend on the configuration enclosed within r.
- **With p as the physical pressure:** the closure acts as a modification of gravity inside matter.

- $\xi = 0$ is forced: outside a sphere, Φ_W has a term in $1/r^4$ that no other potential cancels, and the closure's exterior is exactly Schwarzschild. Monte Carlo check: `phi_w_sfera.py`.
- Matching the remaining coefficients gives $\beta = \gamma = 1$, $\zeta_3 = 0$, $\zeta_2 = 1/2$ and $\zeta_4 = -1$.
- Every theory derived from an invariant action has $\alpha_3 = \zeta_i = 0$ (Lee, Lightman and Ni 1974).
- Binary pulsars bound $|\alpha_3 + \zeta_2| < 4\times10^{-5}$, and hence ζ_2 (Will 2014), and the strong-field $|\zeta_2| < 1.3\times10^{-5}$ (Freire and Wex 2024).
- So, in this reading, the closure is not the static limit of any metric theory derived from an action, nor of any metric theory compatible with these bounds.
- Moreover, by an argument of Will (1976, 2014), in any reasonable theory $6\zeta_4 = 3\alpha_3 + 2\zeta_1 - 3\zeta_3$: $\zeta_4 = -1$ with $\zeta_3 = 0$ would require $3\alpha_3 + 2\zeta_1 = -6$, while $|\alpha_3| < 4\times10^{-20}$.

Causal stars, effective stress (`codice/sforzo_causale.py`).

- **Slopes.** In the maximum-mass closure star with $c_s^2 = 1$, the slopes $dp_r/d\varepsilon$ and $dp_t/d\varepsilon$ of the effective stress do not exceed 0.922 and 0.917. They are not characteristic speeds: the effective medium has no prescribed response to perturbations.
- **Dominant energy condition.** It holds in this star.
- **Trace at the centre of the maximum-mass stars,** closure S/ε against GR $3p_c/\varepsilon_c$:
- $c_s^2 = 1$: 1.52 against 2.01;
- $c_s^2 = 1/2$: 0.90 against 1.13;
- $c_s^2 = 1/3$: 0.65 against 0.79;
- SLy: 1.20 against 1.62;
- APR4: 1.52 against 2.11.

What changes in what is claimed.

- **Density profiles.** In «What is not claimed», «a bound over all density profiles» is now a proved bound, 12/25, for non-increasing profiles. The supremum 0.4662 is a conjecture.
- **Post-Newtonian constraints.** «Nothing on … post-Newtonian constraints» is replaced by the two readings above.
- **Everything else** stands as in version 2.34.1.

Reproduction

Each table is reproduced by one script in `codice/` (Python 3 with numpy, scipy, sympy):

- `sforzo_efficace_342.py`: closed forms, pressures, effective stress;
- `stelle_340.py`: five profiles in the same coordinate;
- `stella_334.py`, `densita_338.py`, `pressione_341.py`: supporting checks;
- `einstein_vuoto_esatto.py`: vacuum;
- `domini_e_masse.py`: two centres, and the Komar–ADM flux.
- `massa_raggio.py`: mass–radius curves with SLy and APR4, GR and closure, and the figure (a few minutes).

Added in version 2.34.1 (run them from `codice/`; the JSON files are written in the current folder):

- `buchdahl_1964_isotropia.py`: Buchdahl's derivation, symbolic and numerical (a few minutes);
- `confronto_put.py`: comparison with the anisotropy law of Puttasiddappa et al. (2026);
- `causale.py`: causal equations of state, turning point, energy conditions;
- `energia_legame.py 2` (2 = number of processes), then `ricontrollo_masse.py`: binding energy, 17 equations of state.

Added in version 2.34.2 (run them from `codice/`; the JSON files are written in the current folder, and programs that need the output of another one read it from the current folder or from `risultati/`):

- `causale_massimo.py`, then `sforzo_causale.py`: causal equations of state, located maximum, effective stress;
- `massa_raggio_stretto.py` (about half an hour with 2 processes), then `figure_articolo.py <folder>`: SLy and APR4 with tight tolerances, and the figures of the manuscript;
- `rumore_raggio.py`: the numerical noise of the radius with the settings of `massa_raggio.py`;
- `legame_massa_esatta.py 1` and `legame_massa_massima.py 1` (1 = number of processes), then `velocita_centro.py`: binding energy at exact mass for the 16 admitted equations of state, and $dp/d\varepsilon$ at the centre;
- `energia_legame_senza_ALF2.py`: the binding-energy relations of version 2.34.1 without ALF2;

- `traccia_sforzo.py`, `ppn_interno.py`, `phi_w_sfera.py`: effective stress (symbolic), post-Newtonian expansion, the potential Φ_W;
- `profili_a_gradini.py 2` and `gradini_ottimizzazione.py`: step profiles.

Computations were carried out with AI assistance; the results before the section on realistic equations of state were re-run by a second AI assistant, not by a human referee (`documenti/VERIFICA_ESTRATTO.md`). Working notes, in Italian, are in `documenti/`.

The author's motivation for the closure is explained in `WHERE_THIS_COMES_FROM.md`; a fuller account of the author's model is in preparation.

References

- H. A. Buchdahl, Phys. Rev. 116, 1027 (1959).
- D. R. Brill, R. W. Lindquist, Phys. Rev. 131, 471 (1963).
- R. L. Bowers, E. P. T. Liang, ApJ 188, 657 (1974).
- R. Beig, Phys. Lett. A 69, 153 (1978).
- L. Herrera, N. O. Santos, Phys. Rep. 286, 53 (1997).
- B. V. Ivanov, Phys. Rev. D 65, 104011 (2002).
- M. Bañados, P. G. Ferreira, Phys. Rev. Lett. 105, 011101 (2010).
- P. Pani, V. Cardoso, T. Delsate, Phys. Rev. Lett. 107, 031101 (2011).
- G. J. Olmo, Int. J. Mod. Phys. D 20, 413 (2011).
- T. Jacobson, Class. Quantum Grav. 24, 5717 (2007).
- H. Andréasson, J. Differential Equations 245, 2243 (2008).
- P. Karageorgis, J. G. Stalker, Class. Quantum Grav. 25, 195021 (2008).
- J. S. Read, B. D. Lackey, B. J. Owen, J. L. Friedman, Phys. Rev. D 79, 124032 (2009).
- D. Choudhury et al., Astrophys. J. Lett. 971, L20 (2024).
- T. Salmi et al., Astrophys. J. 974, 294 (2024).
- A. J. Dittmann et al., Astrophys. J. 974, 295 (2024).

Added in version 2.34.1:

- H. A. Buchdahl, Astrophys. J. 140, 1512 (1964), doi:10.1086/148055.
- M. S. R. Delgaty, K. Lake, Comput. Phys. Commun. 115, 395 (1998), arXiv:gr-qc/9809013.
- T. W. Baumgarte, N. Ó Murchadha, H. P. Pfeiffer, Phys. Rev. D 75, 044009 (2007).
- P. H. Puttasiddappa, N. Bretón, S. E. Perez Bergliaffa, Phys. Rev. D 114, 084009 (2026), arXiv:2608.10991.
- A. Alho, J. Natário, P. Pani, G. Raposo, Phys. Rev. D 106, L041502 (2022).
- S. Koranda, N. Stergioulas, J. L. Friedman, Astrophys. J. 488, 799 (1997).
- J. M. Lattimer, M. Prakash, Astrophys. J. 550, 426 (2001).
- C. Breu, L. Rezzolla, Mon. Not. R. Astron. Soc. 459, 646 (2016).
- P. Podsiadlowski et al., Mon. Not. R. Astron. Soc. 361, 1243 (2005).

Added in version 2.34.2:

- D. L. Lee, A. P. Lightman, W.-T. Ni, Phys. Rev. D 10, 1685 (1974).
- C. M. Will, Astrophys. J. 204, 224 (1976).
- C. M. Will, Living Rev. Relativ. 17, 4 (2014).
- P. C. C. Freire, N. Wex, Living Rev. Relativ. 27, 5 (2024).
- H. Hernández, L. A. Núñez, Can. J. Phys. 82, 29 (2004).
- J. M. Lattimer, M. Prakash, Phys. Rep. 621, 127 (2016).
- M. C. Miller et al., Astrophys. J. Lett. 1000, L48 (2026).